Cail Daley

Observational cosmologist measuring the growth of structure with weak and CMB lensing, through AI agents and the verification that makes their work trustworthy.

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POSITIONS

2024–PRESENT **Postdoctoral Researcher** CEA Paris-Saclay, DAp / UMR AIM, CosmoStat

2019–2024 **Graduate Research Assistant** University of Illinois Urbana-Champaign

EDUCATION

2024 **Ph.D., Astronomy** University of Illinois Urbana-Champaign
Advisor Gil Holder. Thesis: "CMB Lensing Measurements with Two Years of Data from the SPT-3G Survey."

2018 **B.A., Astronomy & Physics** Wesleyan University
Thesis advisor A. Meredith Hughes. High honors.

RESEARCH

2024–PRESENT **Cosmic shear with UNIONS**

- Main postdoc on the first cosmic-shear cosmology from UNIONS, the northern optical survey. Led the B-mode validation (first author, *A&A*, in press) and presented the first cosmological constraints at the Rencontres de Moriond (2026).
- That paper was carried out end to end by AI agents: ∼10,000 lines of agent-written analysis code, figures and manuscript, with my role in the analysis design and the checks that let agents verify their own work.
- Now running the blinded validation of the next, larger UNIONS cosmic-shear analysis, aiming for cosmological results by mid-2027.

2024–PRESENT **Euclid × CMB lensing**

- Leading the Euclid × SPT-3G cross-correlation paper (6×2pt), under the SPT–Euclid Memorandum of Understanding that I also lead; aiming for results by mid-2027.
- Building the end-to-end blinded analysis of Euclid shear and clustering crossed with SPT-3G and ACT CMB lensing, from data vectors and covariance to chains, run as an agent-orchestrated pipeline on the ASTRA analysis specification.

2024–PRESENT **Kinematic lensing**

- Supervising master's students developing symmetry-based kinematic lensing, from methodology to its first application to integral-field survey data.

2019–2024 **CMB lensing with SPT-3G**

- Led one of three independent pipelines in the 2019+2020 SPT-3G lensing analysis, extending the flat-sky quadratic estimator to polarization and 220 GHz with minimum-variance frequency combination and foreground bias-hardening.
- Produced the input maps and point-source masks for the SPT-3G two-year power-spectrum and lensing analyses: 3,286 observations, ∼60,000 CPU hours.

- Cross-validated the deepest CMB lensing maps to date, reaching a ∼2% lensing amplitude with percent-level systematic uncertainties.

2015–2018 **Debris disks**

- Led ALMA observations of the AU Mic debris disk, inferring the mass of stirring bodies from its vertical structure (first author, *ApJ*, 2019).

COLLABORATIONS & ROLES

2024–PRESENT **Euclid Consortium** CMB cross-correlations science working group. Lead, Euclid × SPT-3G cross-correlation paper; lead, SPT–Euclid MoU; global maintainer of the consortium's GitHub organization.

2024–PRESENT **UNIONS** Weak-lensing team; main postdoc on the first cosmic-shear release.

2019–PRESENT **SPT-3G** Collaboration member; lensing pipeline lead for the 2019+2020 analysis.

2025–PRESENT **Lightcone Research** Core team member. Open-source tools for verifiable, AI-mediated science, joint between CNRS and UC Berkeley; developer summit, Berkeley (2026).

PUBLICATIONS

Two first-author papers, substantive individual contributions to ten more, and coauthorship of 38 further collaboration papers, chiefly SPT-3G. 35 refereed; h-index 21; ∼2,100 citations (NASA ADS, October 2026).

2026 **C. Daley** et al., "UNIONS-3500 Weak Lensing: II. B-mode validation for cosmic shear." *A&A*, in press.

2019 **C. Daley** et al., "The Mass of Stirring Bodies in the AU Mic Debris Disk Inferred from Resolved Vertical Structure." *ApJ* 875, 87.

AWARDS & FELLOWSHIPS

2026 CNRS AI Rising Talents finalist, one of five from 42 applicants

2023 APS Division of Astrophysics Student Travel Grant

2021 Chambliss Astronomy Achievement Student Award AAS 238th Meeting

2019 Center for Astrophysical Surveys Graduate Fellowship UIUC, $30,000

2015 NASA Connecticut Space Grant Undergraduate Research Fellowship

SUPERVISION

2025–PRESENT **Serena Nakhle** master's, CosmoStat. First application of symmetry-based kinematic lensing to integral-field survey data (SAMI, WiFeS).

2024–2025 **Jordy Ram** master's, CosmoStat. Kinematic lensing methodology and its first extension to elliptical galaxies; paper in preparation.

2022–2023 **Zimo Qu** undergraduate, UIUC. Stellar flares observed simultaneously by SPT and TESS.

2020–2021 **Society for Equity in Astronomy** Mentor to three undergraduate students.

TALKS

NOV 2026 Colloque national Dark Energy, Paris INVITED · FORTHCOMING
Joint UNIONS cosmology talk with C. Murray

SEP 2026 FORTH Workshop on Physics-Informed Machine Learning, Heraklion KEYNOTE & TUTORIAL
"Scaling Scientific Labor with Agentic Engineering"

MAR 2026 Rencontres de Moriond, Cosmology CONTRIBUTED
"UNIONS-3500: First Cosmological Constraints from 2D Cosmic Shear"

Feb 2026	CosmoStat Learning Slot TUTORIAL SERIES "Agentic AI for Scientific Research"
Jul 2025	TITAN/TOSCA Meeting, European weak-lensing consortium "UNIONS: First Cosmic Shear Constraints from 3000 deg^2 of Northern Sky"
Apr 2025	NRC Herzberg Astronomy & Astrophysics Colloquium, Victoria INVITED "Cosmology with SPT-3G"
Apr 2025	UBC Cosmo Coffee, Vancouver INVITED "Cosmology with SPT-3G"
Mar 2025	56th ESLAB Symposium, Leiden KEYNOTE "Cosmology with SPT-3G"
Jan 2025	Deep CosmoStat Days, CEA Paris-Saclay "SPT Component Separation Challenges"
Jan 2024	AAS 243rd Meeting DISSERTATION TALK "CMB Lensing Measurements with Two Years of Data from the SPT-3G Survey"
Apr 2023	APS April Meeting "CMB Lensing Measurements with SPT-3G"
May 2022	KIPAC Tea Talk, SLAC INVITED "Lensing Maps and Transient Science with the South Pole Telescope"

SERVICE & OUTREACH

Feb 2027	Organiser, SciAI Days, CEA Paris-Saclay departmental workshop on AI for scientific research
2026	Paris Astronomy on Tap public talk, "Astronomy at the End of the World"
2020–2022	Organiser, UIUC Astronomy on Tap monthly public events, livestreamed through the pandemic
2020–2023	SPT First Discoveries science lessons at Fiske Elementary, Chicago
2019	Education Justice Project programming and astronomy-data workshop, Danville Correctional Center

SKILLS

Computing	Python, Julia, Unix, HPC and SLURM, Snakemake, Apptainer, git and CI, LaTeX, web development.
Cosmology	CAMB, CosmoSIS, HEALPix, NaMaster, TreeCorr, lenspyx, emcee; Bayesian inference.
Agents	Claude Code and Codex; multi-agent orchestration; author of felt and Shuttle.
Languages	English (native), French (B2).